

Chedam: work log

Cumulative record of every working day, newest first. This copy covers up to Monday, October 5, 2026. Owner: Sreya.
Baseline: Master Specification v1.0.

Days in this log

- **2026-10-05:** P0 steps 1-3 done, steps 4-6 built and deployed, phones paired: hub base, schema + event log, access, devices, forgot PIN, client app; work log started
- **2026-10-03:** New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed
- **2026-10-02:** P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

P0 steps 1-3 done, steps 4-6 built and deployed, phones paired: hub base, schema + event log, access, devices, forgot PIN, client app; work log started

Summary

- Full health check of the re-flashed Pi: healthy, OS up to date, no hub software yet.
- Restored admin access (the sudoers rule is lost on every re-flash).
- Changed `setup-hub.sh` so the display setting can be switched **both ways** (monitor on or headless) and re-runs leave the boot config unchanged.
- Ran hub setup with `--keep-display`, rebooted and verified everything. HTTPS works by name and by IP.
- iPhone 13 Pro and Android 9 installed the new "Chedam Store CA" and opened the hub page without warnings ("Chedam hub is running").
- Started this cumulative work log (`Work Log/`).
- P0 step 1 (hub base) is complete.
- P0 step 3 (access) is complete: permission engine with role templates, overrides and BR-33, PIN sign-in with lockout, owner recovery code, "Who can do this?". 92 more tests (124 in total) passing; deployed and checked on the Pi.
- P0 step 4 (devices) is built and tested on the PC (67 new tests, 192 in total): pairing by code/QR or by approval request, device key checked on every request, name-then-PIN sign-in on paired devices only, one person per device, device manager (approve, lock, sign out, rename, assign, remove), certificate guide. Deployed to the Pi and checked; pairing the real phones is next.
- Forgot PIN decided and built: a manager sets a temporary PIN, and the person chooses their own at the next sign-in.
- P0 step 5 (client shell) is built and tested on the PC: an installable Svelte app that opens offline, with the connectivity bar, name-then-PIN sign-in, choosing a new PIN, owner sign-in, idle auto-lock, large text and high contrast, and the device manager with QR codes. It replaces the temporary pages. 213 hub tests pass.
- P0 step 6 (setup wizard) is built, tested and deployed: one-time setup code on a new hub, owner and first device in one step, recovery code, business profile, logo and receipt, team, storage, features, references, backups; skipped steps become tasks. 251 hub tests pass.
- Re-tested the step 3 bugs and issues: all fixed. A third copy of one bug (recovery code) was found and fixed.
- P0 step 2 (schema v1 + event log) is complete: 14 tables, reference data, dev sample store, a transactional append-only event log, 32 automated tests passing, deployed to the Pi with a backup first.

Naming note: the local folder `chedam-updates/` in the project is Sreya's YouTube and notes repo (gitignored). It is **not** the update channel. Nothing is renamed; the public update-package repo name will be decided when step 9 (Updates) starts.

1. Health check (before any change)

One SSH session ran these checks:

```
whoami; hostname; hostname -I; cat /proc/device-tree/model; grep PRETTY /etc/os-release; uname -r
uptime; sudo -n true; free -m; df -h / /boot/firmware; lsblk
vcgencmd measure_temp; vcgencmd get_throttled; vcgencmd measure_volts core
timedatectl; ip route; ping -c2 1.1.1.1
apt list --upgradable; dpkg-query -W <packages>; systemctl is-active ...; systemctl --failed
cat /boot/firmware/config.txt /boot/firmware/cmdline.txt
```

Area	Result
Hardware	Pi Zero 2 W Rev 1.0, 33.2 °C, throttled 0x0, core 1.2875 V

OS	Debian 13 trixie, kernel 6.18.50+rpt-rpi-v8, 0 pending updates
Uptime	1 day 12 h, load 0.00, no failed units
Memory / disk	415 MB total, 273 MB available / 119 GB card, 2.3 GB used
Network	192.168.50.101 via 192.168.50.1, internet ping 7 ms
Time	NTP synced, America/Vancouver, no RTC
sudo	Needed a password (re-flash reset it)
Hub software	Not installed: no caddy, chrony, ufw, sqlite3 or <code>/opt/chedam</code>
Display	video=HDMI-A-1:1280x720@60D present (monitor in use)

2. Admin access

Failed case: `Permission denied (publickey)` when restoring sudo

- The `ssh -t chedam "...sudo tee..."` command was typed **inside an SSH session on the Pi** (prompt `chedam@chedam:~ $`).
- So the Pi tried to SSH into itself and had no key for that. Re-flashing was not needed.
- Fix: run only the inner command on the Pi:

```
echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmod 440 /etc/sudo
sudo -n true && echo OK
```

3. Display becomes a switch (DL-24 updated)

Sreya's requirement: the monitor is optional on the Zero 2 W pilot, but the option must stay. Pi 4 production hubs will use a display to operate the hub directly.

Change in `hub/scripts/setup-hub.sh` (boot config section):

- Removes any earlier chedam block, then writes a block between `# chedam-begin` and `# chedam-end`.
- `--keep-display`: display driver `vc4-kms-v3d` and display detection **on**, Bluetooth off.
- No flag: headless. The display driver and detection are off, `gpu_mem=16`, Bluetooth off; this saves ~50 MB of RAM.
- Audio and camera detection are always off.

```
sudo bash ~/hub/scripts/setup-hub.sh --keep-display # monitor on (current)
sudo bash ~/hub/scripts/setup-hub.sh                # headless
sudo reboot                                          # after switching
```

Failed case: an extra `[a11]` line on every run

- The test (switching off, on, off, on on a copy of `config.txt`) worked, but each run appended another `[a11]` line. It was harmless but untidy.
- Fix: moved `[a11]` inside the marked block. The original config (`config.txt.chedam-orig`, saved by the first run) was restored and setup run twice. Both runs gave the same checksum (`d91ff929`), and the only differences from the original were audio off, camera detection off and the chedam block.

4. Hub setup on 192.168.50.101

```
scp -r scripts system chedam:hub/
# on the Pi: download + verify PocketBase
wget pocketbase_0.40.4_linux_arm64.zip checksums.txt
grep linux_arm64 checksums.txt | sha256sum -c - # OK
sudo apt-get update
sudo bash ~/hub/scripts/setup-hub.sh --keep-display
sudo systemctl reboot
```

Verification after reboot:

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon, ufw active; no failed units
Memory	PocketBase 28 MB + Caddy 37 MB = 65 MB (budget 150 MB, NFR-08); 264 MB available with display on
Watchdog	RuntimeWatchdogUsec=14s
Time	chrony, 0.0000007 s from NTP, leap normal
Temperature	38.6 °C, throttled 0x0
Display	/dev/dri/card0 present
Firewall	22, 80, 443/tcp and 5353/udp allowed (IPv4 + IPv6)
CA download	http://chedam.local/ca.crt → 200, "Chedam Store CA - 2026 ECC Root", valid to 2036-08-13
HTTPS	https://chedam.local/api/health and https://192.168.50.101/api/health → 200
Certificate check	openssl s_client ... -CAfile ca.crt → Verify return code 0, with and without SNI
HTTP	Redirects to HTTPS (302)
Caddy errors since boot	0

PC test commands (Git Bash; `--resolve` avoids Windows mDNS problems, and `--ssl-no-revoke` avoids the Schannel revocation error):

```
curl --resolve chedam.local:80:192.168.50.101 -o ca.crt http://chedam.local/ca.crt
curl --ssl-no-revoke --resolve chedam.local:443:192.168.50.101 --cacert ca.crt https://chedam.local/a
curl --ssl-no-revoke --cacert ca.crt https://192.168.50.101/api/health
```

5. Phones (risk R5, first part)

Device	How	Result
iPhone 13 Pro, iOS 26.6.1	Safari <code>http://chedam.local/ca.crt</code> → install profile → Settings > General > About > Certificate Trust Settings → on	<code>https://chedam.local</code> shows the placeholder, no warning
Android 9	Chrome <code>http://192.168.50.101/ca.crt</code> → install for VPN and apps	<code>https://192.168.50.101</code> shows the placeholder, no warning

The camera test over HTTPS comes later, when the client app exists.

6. Work log started

- Folder `Work Log/`: one Markdown source per day in `Work Log/entries/YYYY-MM-DD.md`.
- `tools/worklog/build_worklog.py` builds `Work Log/Chedam-Work-Log-YYYY-MM-DD.pdf` with all days, newest first.
- Earlier PDFs are kept, so each day has its own complete snapshot.

```
python tools/worklog/build_worklog.py # newest day
python tools/worklog/build_worklog.py 2026-10-05 # a specific day
```

The days 2026-10-02 and 2026-10-03 were written up from the session records and `docs/P0-pi-setup-log-2026-10-03.pdf`.

7. P0 step 2: schema v1 and event log

Spec basis: Section 10 (data model, event log), Section 13 (build process: migrations and sample data first), NFR-20 (reversible migrations), BR-30 (UTC), BR-33 (managers cannot grant more than they hold).

Files added

File	Purpose

hub/pb_migrations/1791200001_p0_schema_v1.js	13 new tables + the PocketBase <code>users</code> collection extended: business, locations, settings, modules, permissions, roles, permission_overrides, devices, storage_areas, tasks, events, backups, updates. Common fields on every table. API rules all closed (superuser only) until step 3
hub/ pb_migrations/1791200002_p0_reference_data.js	22 modules (3 core always on, 15 switchable, 4 later), shop-type presets, 20 P0 permissions (5 owner-only), 5 role templates, 12 default settings
hub/ pb_migrations_dev/1791200100_dev_sample_store.js	Dev-only sample store: "Demo Grocery", main location, grocery preset modules, 4 storage areas, 5 people (one per role, random unknown passwords), 1 setup task
hub/pb_hooks/event_log.pb.js + lib/event_log.js	Event log: one <code>events</code> row per create/update/delete, in the same transaction; actor and device stamped from the request; append-only
hub/tests/step2-schema-eventlog.test.mjs	32 automated tests against a throwaway local PocketBase
hub/scripts/deploy.sh	PC → hub deploy: copy, SQLite online backup + integrity check, rsync, restart, health, migrations, errors
docs/conventions.md	Data, event-log, migration and workflow rules for all phases

Design choices

- **Common fields:** `id` (a device may create it offline), `created_at`, `updated_at`, `created_by`, `updated_by`, `device_id`, `deleted_at`. `updated_by` was added beyond the spec's list, so each change records who made it.
- **The event log runs inside the same transaction** as the change (`onRecord*Execute + runInTransaction`). If the event can't be written, the change is rolled back, so the log never misses a write.
- **Actor and device are stamped by the hub** in request hooks. A client cannot forge `created_by`, and the device comes from the `X-Chedam-Device` header (verified in step 4).
- **Hidden fields** (`pin_hash`, `recovery_code_hash`, `key_hash`, `pairing_code_hash`) never reach the API or the log.
- **Role levels:** owner 100, manager 80, accountant 50, cashier 40, staff 30. Owner-only permissions: `modules.manage`, `setup.run`, `owner.manage`, `backups.restore`, `updates.install`.
- **Sample data is separate** from real migrations, so a store hub never receives demo records.

Commands

```
# local tests (PC, Git Bash)
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/step2-schema-eventlog.test.mjs
# deploy to the Pi with sample data
bash hub/scripts/deploy.sh --sample-data
# verify on the Pi
ssh chedam 'sudo -u chedam-hub sqlite3 /opt/chedam/pb_data/data.db "select table_name, action, count('
```

Failed cases and fixes

Symptom	Cause	Fix
pocketbase.exe: command not found in Git Bash	That shell started before <code>C:\Tools</code> was added to PATH	Full path <code>C:\Tools\pocketbase.exe / PB_BIN</code>
migrate down hung	It waits for a y/N confirmation	Pipe <code>y</code> (input: <code>"y\n"</code> in the tests)
Python check query error (ESCAPE expression must be a single character)	Backslash escaping in an inline script	Filter with <code>substr(name,1,1) != '_'</code>
Sample migration: Cannot read property 'forEach' of undefined	A JSON field read with <code>get()</code> returns raw bytes	<code>JSON.parse(record.getString("value"))</code>

Test stop step hung	Waited for the exit of a process that had already exited	Check <code>exitCode/signalCode</code> first
Create returned 400 "Something went wrong"	<code>e.request</code> doesn't exist in PocketBase 0.40 JS hooks (found with <code>serve --dev</code> logs)	<code>e.requestInfo().headers["x_chedam_device"]</code> (names lowercased, <code>_</code> → <code>_</code>)
A Python edit put real newlines inside JS strings	Escaping in a heredoc	Fixed with a direct edit

Test results (PC, 32/32 passed)

- Schema/reference: 22 modules, 3 core on, 15 switchable, 20 permissions, 5 roles, owner holds all, manager holds no owner-only permission, 12 settings.
- Sample: business + location, 4 storage areas, 5 people, 1 task, 7 grocery modules on.
- Event log: device-made id accepted; forged `created_by` overwritten; device stamped; create/update/soft-delete/delete each logged with correct before/after/changed; delete keeps the last state; no PIN hash, password or tokenKey in the log; events can't be edited or deleted, even by a superuser.
- Atomicity: with the event write forced to fail, the create fails and the record is not in the table.
- Migrations: down 3 and up again both clean.

On the Pi (after deploy)

Check	Result
Pre-deploy backup	<code>pre-deploy-<UTC>.db</code> , integrity ok
Migrations applied	schema v1, reference data, dev sample store
Rows	modules 22, permissions 20, roles 5, settings 12, business 1, locations 1, storage_areas 4, users 5, tasks 1
Events	78 = every write above (incl. 7 module updates for the grocery preset)
Secrets in events	0
SQLite journal	WAL
PocketBase memory	42 MB (budget 150 MB)
Errors in the journal	none

8. P0 step 3: Access

Spec basis: FR-1.03 (owner password, PIN, printed recovery code), FR-1.04 (people with role and PIN), FR-1.09 (owner signs in on any device), FR-1.10 (role templates, per-person overrides with end date, "Who can do this?"), NFR-11 (hub-side permission checks, PIN lockout after 5 tries, auto-lock), BR-33 (managers can't grant more than they hold).

Files added or changed

File	Purpose
<code>hub/pb_migrations/1791200003_p0_access.js</code>	PIN and recovery code become bcrypt password fields (hidden); <code>pin_set</code> and <code>recovery_code_created_at</code> added; API rules set on all 14 tables; token lifetime 12 h. Reversible
<code>hub/pb_hooks/lib/access.js</code>	Permission engine: effective permissions (role + allow overrides – deny overrides; owner = all), the per-table access map, protected fields, self/assignee field limits, BR-33 checks for people, roles, overrides, modules, settings, business
<code>hub/pb_hooks/lib/auth.js</code>	PIN rules, lockout counter, recovery code generator, secret check
<code>hub/pb_hooks/access.pb.js</code>	Guard on every list/view/create/update/delete; password sign-in lockout; token refresh check; endpoints (below)

hub/ pb_migrations_dev/1791200101_dev_sample_pins.js	Dev-only PINs for the 5 sample people (test values; dev hubs only)
hub/pb_hooks/lib/event_log.js	Fix: put the original app back after the transaction (see failed cases)
hub/tests/lib/hub.mjs, step3-access.test.mjs, run-all.mjs	Shared test harness, 92 access tests, one-command runner
docs/conventions.md, docs/P0-mini-spec.md	Access rules for later phases; decisions DL-28 to DL-32

Endpoints

Method and path	Who	What
GET /api/chedam/auth/pin-users	anyone on the LAN (step 4: paired devices only)	Names for the till sign-in screen
POST /api/chedam/auth/pin	anyone	{user, pin} → sign-in token; wrong tries count toward logout
POST /api/chedam/users/{id}/pin	yourself, or users.manage above them	Set or change a PIN (4-6 digits, no repeats or straight runs)
POST /api/chedam/users/{id}/unlock	users.manage above them	Clear a logout
POST /api/chedam/owner/recovery-code	owner	New printable code, shown once (format XXXXX-XXXXX-XXXXX-XXXXX)
POST /api/chedam/auth/recover	anyone with the code	{email, code, new_password} → new password + sign-in; the code is used up
GET /api/chedam/access/me	signed in	My role and effective permissions
GET /api/chedam/access/users/{id}	self or users.view	Someone's effective permissions and where each comes from
GET /api/chedam/access/who-can/{code}	users.view	"Who can do this?" (via role or override, with end date)

Who may do what (summary of the access map)

- **Anyone signed in:** reads the business profile, locations, settings, modules, permissions, roles and storage areas.
- **People:** list with users.view; a person can view and edit their own record (name, email, phone, language, accessibility, password only); add/edit others needs users.manage; giving or changing roles needs access.manage; no hard delete.
- **Overrides:** access.manage only; never on yourself; only on people below you; never grant what you don't hold; owner-only permissions only by the owner.
- **Roles:** templates can't be deleted; only the owner edits templates; the owner role always holds everything; managers create and edit only roles below their own level.
- **Modules:** owner only; only enabled can change; core can't be switched off; "later" modules can't be switched on; who/when stamped.
- **Settings:** security.* owner only; keys can't be renamed.
- **Audit log:** events.view. Tasks: tasks.view / tasks.manage (the assignee may update status/note). Devices: devices.view / devices.manage. Backups and updates: read-only through the API.

Failed cases and fixes

Symptom	Cause	Fix
No way to check a bcrypt PIN in JS hooks	\$security has no bcrypt	Use a PocketBase password field: record.getRaw("pin").validate(plain)

Owner recovery sign-in failed: failed to delete all previous MFAs ... transaction has already been committed	The event-log wrapper replaced <code>e.app</code> with the transaction and never restored it, so PocketBase's clean-up after a password change used a finished transaction. This would have broken every password change.	Restore <code>e.app</code> in a <code>finally</code> block after <code>runInTransaction</code>
5th wrong password returned 400, not 423	The failure was counted, but that same response didn't report the new lock	After counting, re-read the person and answer "locked" (same as PIN sign-in)
Assumed PocketBase leaves <code>e.record</code> empty on a wrong password	Traced with debug lines: the hook gets the person and <code>e.next()</code> checks the password	Removed the extra lookup and the debug lines
Step 2 test still sent <code>pin_hash</code> and reversed only 3 migrations	The schema changed in step 3	Test uses the <code>pin</code> field, checks for no <code>\$2a\$</code> hashes, and reverses every Chedam migration it finds
<code>sed</code> edits to the test harness didn't apply	Escaping in shell quoting	Direct edits

Test results (PC)

`node hub/tests/run-all.mjs` → step2 32/32, step3 92/92.

Step 3 covers: - PIN sign-in, the PIN list showing no secrets, and wrong PINs refused. - Lockout after 5 wrong PINs and after 5 wrong passwords (423), and unlock by a manager but not a cashier. - The PIN rules. - Who can read and change each table. - Managers adding cashiers but not owners or other managers, and editing people below them but not the owner. - Role changes, and the rule that you can't change your own role. - Suspending someone signs them out at once. - Overrides: allow, deny, past and future end dates, owner-only, and "can't grant what you don't hold". - Role templates, the owner role, and deleting custom roles. - The module, setting and business rules. - The recovery code: shown once, case-insensitive, used once. - Who-can and my-permissions. - The audit trail: lockouts, overrides and module switches are logged, and no hashes appear anywhere.

On the Pi (after deploy, checked on the Pi itself)

Check	Result
Deploy	Backup + integrity ok; migrations <code>1791200003_p0_access.js</code> and <code>1791200101_dev_sample_pins.js</code> applied; no errors
PIN list	5 people
Signed-out request	401
PIN sign-in time	0.50 s (bcrypt cost 10 on the Pi Zero 2 W)
<code>access/me</code> for the cashier	role cashier, permissions <code>tasks.view</code>
Cashier lists people	403
bcrypt hashes in the event log	0
PocketBase memory	44 MB

The dev PINs are in the dev seed file only and are not repeated in this log.

9. Re-test of the step 3 bugs and issues

Sreya asked which 2 bugs and 2 issues came up in step 3, whether they were fixed, and to test them again.

#	Kind	What	Status after re-test
1	Bug	Event-log wrapper left <code>e.app</code> pointing at a finished transaction, so the owner's recovery sign-in (and every password change) failed	Fixed. Tests "recovery code + new password signs the owner in" and "new password works" pass
2	Bug	The 5th wrong password answered 400 instead of 423 (locked)	Fixed. Test "5 wrong passwords lock the owner too" passes

3	Issue	PocketBase JS hooks have no bcrypt function	Solved with PocketBase <code>password</code> fields; PIN tests pass
4	Issue	The step 2 test was out of date after the step 3 schema change	Updated; step 2 suite passes (now 33 tests)
5	Bug found now	Same as bug 2 on the recovery code path: the 5th wrong code answered 400, not 423	Fixed: one shared <code>auth.failAndThrow()</code> now serves PIN, password and recovery code. New test "5th wrong recovery code answers 423"

Baseline before any change: `node hub/tests/run-all.mjs` → 32/32 and 92/92.

10. P0 step 4: Devices (FR-1.07, 1.08)

Files added or changed

File	Purpose
hub/ pb_migrations/1791200004_p0_devices.js	paired_via, paired_at on devices; generic create closed; settings devices.pairing_code_minutes (10), devices.online_seconds (120), devices.max_pending_requests (10). Reversible
hub/pb_hooks/lib/devices.js	Device check for every <code>/api/</code> request, pairing codes, keys, last seen, device view
hub/pb_hooks/devices.pb.js	Middleware + endpoints (below)
hub/pb_hooks/access.pb.js	Password and PIN sign-in need a paired device (owner excepted); sign-in records who is on the device; assigned devices; recovery-code fix
hub/pb_hooks/lib/access.js	Devices: generic create closed, protected fields, BR-33 on assigning
hub/pb_hooks/lib/auth.js, lib/event_log.js	<code>failAndThrow()</code> ; device stamped only when verified
hub/pb_public/	Temporary pages until the Svelte client (step 5): <code>device-setup.html</code> (certificate guide, works over HTTP), <code>pair.html</code> (pair, then pick your name and enter PIN), <code>devices.html</code> (device manager with QR code), <code>assets/</code> (shared CSS/JS, vendored QR library, <code>VENDORED.md</code>)
hub/system/Caddyfile	HTTP serves the guide and its CSS; <code>/hub.json</code> gives the hub IP for QR codes
hub/scripts/deploy.sh	Also updates the Caddy config when it changed (keeps the IP, validates, keeps a copy, reloads)
hub/tests/lib/hub.mjs, step2..., step3..., step4-devices.test.mjs	<code>pair()</code> helper and automatic device headers; existing suites now sign in on paired devices; 67 new tests
docs/P0-mini-spec.md, docs/conventions.md	DL-33 to DL-38, device rules, the open "forgot PIN" item

Endpoints

Method and path	Who	What
POST <code>/api/chedam/devices/pairing-code</code>	devices.manage	{name, type} → one-time code XXXX-XXXX (10 min)
POST <code>/api/chedam/devices/pair</code>	new device	{code} → device id + key, approved
POST <code>/api/chedam/devices/request</code>	new device	{name, type} → id + key, waits for approval
GET <code>/api/chedam/devices/me</code>	the device	Its status and who is signed in (works while pending or locked)
POST <code>/api/chedam/devices/me/sign-out</code>	the device	Sign out whoever is on it
GET <code>/api/chedam/devices</code>	devices.view	Device manager list: online, signed in, assigned, version, last seen, browser
POST <code>/api/chedam/devices/{id}/approve, lock, unlock, sign-out, revoke</code>	devices.manage	Device manager actions
PATCH <code>/api/collections/devices/records/{id}</code>	devices.manage	Rename, type, assign person or printer

Sign-in flow (Sreya: "select name and then PIN")

The device is paired once. After that the sign-in page shows the people's names as large buttons. Tap your name, enter your PIN on the keypad and press OK. A wrong PIN shakes the keypad and shows "Wrong name or PIN". After 5 wrong tries the person is locked (423). "Not you? Back to names" returns to the list. The owner also has "sign in with email and password", which works on any device. "Forgot your PIN?" shows the current options (see the open item in the mini-spec).

Commands

```
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/run-all.mjs
npm pack qrcode-generator@2.0.4      # vendored QR library; integrity checked against the npm registry
```

Failed cases and fixes

Symptom	Cause	Fix
Shell edit script stopped with "unexpected EOF while looking for matching quote"	Quotes and backticks of JS code inside a bash heredoc	Wrote the Python edit script to a file and ran it
Helpers at the top of <code>devices.pb.js</code> would be undefined in handlers	Each PocketBase JS handler runs in its own VM and cannot see top-level variables of the <code>.pb.js</code> file	Moved them into <code>lib/devices.js</code> (loaded with <code>require</code> inside each handler)
Test "key and code hashes never in API output" failed	The test read the record as superuser , and PocketBase shows hidden fields to superusers on purpose	The test now reads as the manager (who never sees them); superuser = development admin only
"Paired as ..." message vanished at once	The status refresh after pairing cleared the message	Show the message after the refresh
PIN typed by the browser test tool did nothing	The tool inserts text without key events (a real keyboard sends them)	Tested by tapping the on-screen keypad, as on a phone
Deploy to the Pi blocked	Deploying to the hub needs Sreya's approval in this session	Sreya approved; deployed

Test results (PC)

`node hub/tests/run-all.mjs` → step2 **33/33**, step3 **92/92**, step4 **67/67** (192 in total).

Step 4 covers: - **Pairing codes**: who may make them, the format, lower case with a space accepted, single use, expiry, approved by the code's maker. - **Device keys**: unknown, wrong or missing keys refused (401); keys never in API output or the log. - **Ask to join**: pending until approved; the cap on open requests (429). - **Device needed**: no PIN list or PIN sign-in without a device; a cashier's password sign-in or token without a device refused; the owner works on an unpaired browser. - **One person per device**: the earlier token is refused, tokens can't move devices, sign-out. - **Device manager**: list (online, signed in, version, own device marked); sign-out, lock (423)/unlock, rename; protected fields; no generic create. - **Assigned devices**: list and sign-in limited; BR-33 on assign and remove. - **Revoke**: device and token refused, no double remove. - **Recovery**: the 423 fix, and owner recovery on a till. - **Log**: last-seen writes throttled (≤1 event for 10 requests), lock/unlock/revoke logged with actor and the manager's device, no hashes. - **Guessing**: 20 wrong codes pause pairing (429).

Browser check (local dev hub, PC)

Paired with a code from the URL. The name list showed 5 sample people. A wrong PIN gave "Wrong name or PIN" and the right PIN signed in (manager), with the Device manager link shown. The device manager showed the till online with its version and browser. Pair a new device showed the QR code, the code and a countdown. At phone width (375 px) the certificate guide opened the Android steps automatically, with no sideways scrolling.

On the Pi (after deploy, checked from the PC over the LAN)

```
bash hub/scripts/deploy.sh --sample-data
```

Check	Result
Deploy	Backup integrity ok; <code>1791200004_p0_devices.js</code> applied; Caddy config updated for 192.168.50.101 and reloaded (old copy in <code>backups/</code>); no errors

Certificate guide over plain HTTP	<code>http://192.168.50.101/device-setup.html</code> 200, its CSS 200; other HTTP pages still redirect to HTTPS (302)
<code>https://192.168.50.101/hub.json</code>	<code>{"ip":"192.168.50.101","name":"chedam.local"}</code>
Pages over HTTPS	<code>pair.html</code> , <code>devices.html</code> , <code>assets/qrcode.js</code> 200
PIN list without a device	403
Made-up device header	401
Wrong pairing code	400
Device settings	3 devices, * settings present
PocketBase memory	40 MB (budget 150 MB)
Temperature	37.0 °C

11. Forgot PIN: temporary PIN (Sreya's decision)

Sreya: "manager reset temporary and user need to reset again".

- A PIN set by someone else (manager with users.manage above the person, or superuser) is **temporary**. It works for 24 h (`security.temp_pin_hours`).
- With a temporary PIN, the hub allows only these until a new PIN is chosen: setting your own PIN, `access/me`, `devices/me` and sign-out. Everything else answers 403 "Choose your own new PIN first." (`auth.pinChangeGate`, run in the request middleware).
- The new PIN must differ from the temporary one, and the PIN rules apply. An expired temporary PIN is refused with a clear message.
- `pin_must_change` and `pin_temp_expires_at` are protected fields, and both changes are in the audit log.
- Migration `1791200005_p0_temp_pin.js` (reversible). Tests: `step3b-temp-pin.test.mjs`, 14/14.
- In the app: the manager uses "Someone forgot their PIN" on the home screen; the person then sees "Choose your own PIN" (enter twice).

12. P0 step 5: client shell (FR-1.09, FR-12.01, NFR-16/17)

Hub side

File	Purpose
<code>hub/pb_migrations/1791200006_p0_connectivity.js</code>	Settings <code>network.check_url</code> (Cloudflare's 204 check) and <code>network.hotspot_ssids</code>
<code>hub/pb_hooks/lib/net.js</code> , <code>status.pb.js</code>	Check once a minute (cron); GET <code>/api/chedam/status</code> → <code>{internet: offline/router/hotspot, hub_time}</code> ; open to waiting/locked devices too
<code>hub/pb_hooks/access.pb.js</code>	<code>access/me</code> also returns <code>pin_must_change</code> , large text, high contrast, language
<code>hub/tests/step5-status.test.mjs</code>	7 tests: open, values, hub clock, offline/online via the check address, guests can't force a check, waiting device

Client (`client/`)

Part	What
Stack	Svelte 5.57.1, Tailwind 4.3.3, Vite 8.3.2, qrcode-generator 2.0.4 (same integrity hash as checked in step 4). Exact versions; <code>npm audit</code> 0 vulnerabilities
Build	<code>npm run build</code> → <code>hub/pb_public/</code> (about 35 KB gzipped JS + 4 KB CSS). <code>deploy.sh</code> builds it before copying
PWA	Manifest, icons (drawn by <code>client/scripts/make-icons.mjs</code> , no image library needed), service worker generated with the exact file list

Screens	Pair (code / ask to join), Waiting (approval or locked), Names, PIN pad, Choose your own PIN, Owner sign-in, Home (modules switched on, "Someone forgot their PIN", My settings, Install), Devices (manager)
Connectivity bar	Online (router) / Online (hotspot) / Offline, "Hub not reachable", device clock warning (2+ min)
Auto-lock	After <code>security.auto_lock_minutes</code> idle: signed out on that device, name list returns
Removed	The temporary step 4 pages (<code>pair.html</code> , <code>devices.html</code> , <code>assets/chedam.js</code> , vendored <code>qrcode.js</code>). The certificate guide moved to <code>client/public/</code> and now opens the app with <code>/?pair=CODE</code>
Caddy	HTTP serves <code>/device-setup.html</code> and <code>/setup.css</code>

Browser check (local dev hub on the PC)

Check	Result
Session kept	The earlier paired till and Mira's session came back after the app update
Connectivity bar	"Online (router)"
Forgot PIN, end to end	Mira set a temporary PIN for Sam → Sam signed in with it → forced to "Choose your own PIN". A mismatch was caught, reusing the temporary PIN was refused, the new PIN was set and Sam reached his home screen (no Devices tile: staff)
Offline shell	Service worker active, 8 files cached. With the hub stopped , a reload still opened the app with the red "Hub not reachable" bar and a clear message
Recovery	Hub started again → bar back to "Online (router)"; Sam's session came back
Device manager	Mira → Devices → Pair a new device → QR code + code + countdown
Phone size, dark mode	Readable, no sideways scrolling
Certificate guide	Page and <code>setup.css</code> 200; continues to <code>/?pair=CODE</code>

Failed cases and fixes

Symptom	Cause	Fix
Step 2 test "12 default settings" failed (13)	It counted settings; step 4/5 add their own	Checks the 12 step-2 keys by name
Svelte warning: <code>pad</code> not declared with <code>\$state</code>	Component reference in a plain variable	<code>let pad = \$state()</code>
Offline reload showed a blank page	The module script is requested with an <code>Origin</code> header and the hub answers <code>Vary: Origin</code> , so the cached copy (stored without <code>Origin</code>) did not match	Cache look-ups use <code>ignoreVary: true</code> (our files are content-hashed)
Crash "Cannot read properties of null (reading 'user')" when signing out	Home still drew <code>s.me.user</code> for a moment after sign-out cleared it	Home renders only while someone is signed in
After the hub came back, the name list kept its error until the next 30 s poll	Nothing reacted to the hub returning	The app refreshes as soon as the hub answers again; the name list retries
First page load after restarting the dev hub was refused by the browser pane	The page was opened before the server was listening	Waited 2 s and opened it again

Test results (PC)

`node hub/tests/run-all.mjs` → step2 33/33, step3 92/92, step3b 14/14, step4 67/67, step5 7/7 (213).

On the Pi (after deploy, Sreya's go-ahead)

Check	Result
Deploy	App built (35 KB gzipped JS); backup integrity ok; migrations <code>...05_p0_temp_pin, ...06_p0_connectivity</code> applied; Caddy updated and reloaded; no errors

App over HTTPS (IP and chedam.local)	/, sw.js, manifest.webmanifest, icons: 200
Certificate guide over HTTP	device-setup.html, setup.css: 200
/api/chedam/status	internet: router; hub clock matches
Wi-Fi name readable by the hub (nmcli)	yes, so hotspot detection works
PocketBase memory	40 MB

13. Phones paired; bug: the app froze on the iPhone with the Pi off

Sreya paired the Android 9 phone ("9lite") and the iPhone; both appear in the device list. With the Pi powered off, the app on the iPhone froze: no button did anything.

Cause	When the hub is switched off, a phone's requests are not refused; they wait (nothing answers at that address). (1) The service worker waited for page loads with no time limit . (2) Each button waited up to 8 s for the hub, some twice in a row, with no feedback. (3) The status bar checked only every 30 s. The PC test had missed it: a stopped server on the same machine refuses at once
Fix	Page loads get 3 s, then the cached app opens. Requests get 6 s; once the hub is known to be down, requests fail at once until it answers again. The bar checks every 5 s while the hub is down. Signing out with the hub off returns to the name list. "Check" shows "Checking..."
Test	A test server that accepts connections and never answers (like a switched-off hub seen from a phone): bar red after 4.0 s; sign-out 51 ms; reload shows the app at once with the red bar; real hub started again (page not touched): back to "Online (router)" with the name list after 4.3 s
Deployed	Commit <code>6ede814</code> ; deploy ok (backup integrity ok, no errors)

Note: a phone gets the fixed version the next time it opens the app **while the Pi is on**. Open it, wait a few seconds, then close and reopen once.

14. P0 step 6: setup wizard (FR-1.01-1.06, 1.13, 1.15)

Sreya chose: **a setup code on the hub** for the first device (DL-44).

Hub

File	Purpose
hub/ pb_migrations/1791200007_p0_setup.js	devices.paired_via gains "setup" (reversible)
hub/pb_hooks/lib/setup.js, setup.pb.js	Setup code (made at startup, by a once-a-minute job and on demand; kept in pb_data/setup-code); monitor message pb_data/console.issue: GET /api/chedam/setup/status, POST /api/chedam/setup/start, POST /api/chedam/setup/step
hub/pb_hooks/lib/access.js	business.setup_state changes only through the step endpoint
hub/system/chedam-console.path + .service	When the hub's message changes, the monitor's login screen re-reads it (agetty --reload); /etc/issue.d/chedam.issue links to the message
hub/scripts/setup-hub.sh, deploy.sh	Install the monitor units and link; print the setup code on a new hub
hub/tests/step6-setup.test.mjs	38 tests on a hub without sample data

The 38 tests cover: - **The code:** made on a new hub, kept across restarts, shown in the monitor message. - **Refused at start:** a wrong code, a short password, a guessable PIN and a bad email. - **Start:** returns the device key and recovery code once. Afterwards the code file is gone, the monitor message changes and the code can't be reused. - **The new owner:** PIN, password and recovery code all work. The first device is approved and the owner is signed in on it. - **Store records:** one business with Canadian defaults, plus the main location. - **Steps:** - Language, clock and owner are already done after start. - The profile can be edited, but setup_state can't be changed directly. Unknown steps are refused. - A skipped step becomes one task (skipping twice doesn't make two), and doing it later closes the task. When all steps are done, setup is complete. - **Who may run it:** a manager can't mark steps, and start is refused once the store is set up. A PIN the owner sets for someone is temporary. - **Audit log:** setup writes are logged as system:setup, with no secrets.

App

Screen	What
Set up your store (3 parts)	Welcome + language + setup code; clock check against the hub (warns at 2+ minutes); owner name, email, password twice, PIN twice, device name
Recovery code	Shown once, Print button, "I have printed it or written it down" required
Wizard	Progress chips (✓ done, ↻ skipped), resumes at the first unfinished step, every step has Save or Skip
Business profile	Store and legal name, type, address, phone, email, GST/HST, PST, BN, CAD, time zone, fiscal year start, prices with or without tax; the main location gets the same address
Logo and receipt	Logo upload (up to 2 MB), colours from the logo (editable, readability warning), receipt header and footer, live receipt preview
Your team	Add people with role, first PIN (temporary), phone, email
Storage areas	"Add the usual four", add with temperature range, mark not used
Features	Ticked from the shop-type preset, change any; core always on, later ones not available
References	Bank name, processor and merchant ID, CRA payroll, CBSA importer, accountant; warning: no account numbers or passwords
Backups	Explained; skip only (adds the task) until step 7
Home	"Finish setting up your store (N steps left)", task list with "Do it", Store setup tile, "Print a new recovery code" (owner)

Browser check (brand-new local hub, no sample data)

Check	Result
Hub start	<code>setup-code</code> and the monitor message made at startup
First screen	"Set up your store · 1 of 3"; no sideways scrolling at phone width
Wrong code	Refused at the end with a clear message, back to part 1
Clock check	"OK (under 2 minutes)"
Right code typed in lower case without the dash	Owner created; this browser paired as "Owner's computer"; recovery code shown once; Continue locked until confirmed
Business profile	Saved; wizard moved to Logo and receipt (4 of 10)
Logo	Test logo in orange <code>#e8701a</code> and navy <code>#1b2f5e</code> : both found exactly; button colour darkened to <code>#b45714</code> for readable white text; receipt preview with logo, name, address, GST number, footer
Team	Cashier added with a temporary first PIN
Features	Convenience preset ticked Promotions, Labels, Vendors and Purchasing, Regulated Items, Operations
Skips	References and Backups skipped → 8 of 10 done, two tasks on Home with "Do it"
Saved data	Logo file, colours, receipt text, GST number in the database; logo loads without sign-in (200 image/png)

Failed cases and fixes

Symptom	Cause	Fix
Setup routes answered 404	<code>onServe</code> does not exist in PocketBase JS hooks, so the whole file failed to load	<code>onBootstrap</code> (after start-up) plus a once-a-minute job and on-demand from the status endpoint
A <code>sed</code> edit broke a line in <code>setup.js</code>	The inserted comment swallowed the rest of the line	Fixed by hand
Heredoc test script failed in bash	Quotes inside the heredoc	Wrote the test file directly
Svelte warnings "captures the initial value of business"	The forms are meant to start from the saved values once	<code>untrack()</code> makes it explicit

Storage: Shelf showed "0 to 0 °C" and the four areas were in A-Z order	PocketBase number fields have no "empty" (they save 0); all four got the same sort number	0 to 0 shows as "room temperature"; each area gets its own sort number
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On the Pi (after deploy)

Check	Result
Deploy	Backup integrity ok; 1791200007_p0_setup.js applied; monitor units installed; no errors
Monitor message	console.issue = "Chedam hub: https://4" (this dev hub has an owner, so no setup code); /etc/issue.d/chedam.issue linked; chedam-console.path active
/api/chedam/setup/status	needs_setup: false
PocketBase memory	41 MB

The setup code on the **monitor** will be seen for real on a brand-new hub, for example the spare-card restore test in step 7/10. The dev hub keeps its sample store.

Next

1. Sreya: glance at the hub's monitor (login screen shows "Chedam hub: https://192.168.50.101"); on the phones, open the app once with the Pi on to get the new version; as owner, open **Store setup** to try the wizard on the sample store.
2. Phones: sign in with name + PIN on the iPhone and Android 9, install the app, and try it with the Pi off (the freeze fix). PC sign-in completes "3 devices log in".
3. P0 step 7: backups (USB drive, schedule, first backup from the wizard, restore to a spare card).

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2026-10-03 (Saturday, October 3, 2026)

New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed

Source: docs/P0-pi-setup-log-2026-10-03.pdf (detailed log of this day) and docs/P0-restart-new-network.md (the plan). This entry carries its content forward.

Summary

- The TELUS modem gives no admin access, so DHCP reservations are impossible. The dev hub moved behind our **own TP-Link router**, which is LAN-cabled to the bedroom wall port.
- Network: subnet 192.168.50.0/24, gateway 192.168.50.1, 2.4 GHz SSID TP-LINK_2CBE.
- Pi chedam at 192.168.50.101 (MAC 88:A2:9E:5C:17:65, reservation enabled); dev PC DESKTOP-AHG4TD7 at 192.168.50.100.
- The SD card was wiped and re-flashed. Key-only SSH works, and the OS is updated.
- HDMI console made visible. setup-hub.sh gained the --keep-display option.
- Commits: ea0a7e6, 678eee4, bacf585.

Why SSH keys instead of a password

With a key pair, the private key never leaves the PC and the Pi holds only the public key. No secret crosses the network, keys cannot be guessed, and scripts can log in, so password login is switched off.

File	Where	Rule
id_ed25519	PC C:\Users\Venkata\.ssh\	Private. Never share, paste or commit; back it up privately
id_ed25519.pub	PC, same folder	Public, one line ending chedam-dev. Safe to copy
authorized_keys	Pi /home/chedam/.ssh/	Allowed public keys; folder 700, file 600
known_hosts	PC .ssh\	Remembered Pi identities; ssh-keygen -R <ip> after a re-flash
config	PC .ssh\	Alias ssh chedam

Key procedures recorded:

- create the key pair
- put the public key on the Pi with Imager (REMOVE old keys, then BROWSE to the .pub file)
- add or replace keys with type new.pub | ssh chedam "cat >> ~/.ssh/authorized_keys"
- what to do when locked out (monitor + keyboard, or re-flash)

Steps

1. Backed up .ssh with Copy-Item \$env:USERPROFILE\.ssh \$env:USERPROFILE\.ssh_backup -Recurse.
2. Cleared old host fingerprints by deleting known_hosts and known_hosts.old. There was a backup.
3. Router: changed the admin password, set the LAN to 192.168.50.1/24 (different from TELUS 192.168.1.x), kept it in router mode, and turned on 2.4 GHz.
4. Re-flashed the card. Imager: Pi Zero 2 W, Lite 64-bit, chedam/chedam, new Wi-Fi, CA, SSH public-key only.
5. Found the Pi in the router's DHCP client list and created the reservation.
6. Updated the SSH config to HostName 192.168.50.101 and removed HostKeyAlias.
7. Ran sudo apt update && sudo apt full-upgrade -y and checked that sshd -T shows passwordauthentication no.
8. Readiness check: model, OS, IP, active Wi-Fi, multi-user.target; reboot test passed.
9. Forced HDMI output (see below).

Failed cases and fixes

Attempt / symptom	Cause	Fix
ssh-keygen -R <old-ip> errored	PowerShell treats < as an operator	Use real addresses; deleting known_hosts avoided needing the old IP
Key pair deleted during "start fresh"	Not needed: a key is not tied to a network	Restored from .ssh_backup
ssh 192.168.50.101 denied as venkata	No user given, so the Windows name was used	ssh chedam@192.168.50.101
ssh chedam@chedam timed out	Windows cannot resolve a bare name	IP, chedam.local, or the alias
-i \$env:venkata\.ssh\... not found	\$env:USERPROFILE is a real variable, not a placeholder	Keep \$env:USERPROFILE
Permission denied (publickey)	Imager wrote an old remembered key (... COL3o srey...), not the PC's (... LuhpI0dY chedam-dev)	Re-flash; REMOVE old keys, BROWSE to id_ed25519.pub
A command pasted into Imager's key box	Get-Content ... \ Set-Clipboard text pasted as the key	REMOVE, then BROWSE to the file
"REMOTE HOST IDENTIFICATION HAS CHANGED"	Expected after a re-flash (new host key)	ssh-keygen -R 192.168.50.101
Reservation saved as Disabled (TP-Link)	Router default	Tick it, then Enable Selected; leave ARP binding off
HDMI blank	dmesg: vc4-drm: Cannot find any crtc or sizes	Append video=HDMI-A-1:1280x720@60D to the single line in /boot/firmware/cmdline.txt, then reboot
Lines after sudo cp lost	The sudo password prompt ate the pasted lines	Paste one command at a time when sudo may prompt

HDMI fix commands:

```
sudo cp /boot/firmware/cmdline.txt /boot/firmware/cmdline.txt.bak
sudo sed -i '1 s/$/ video=HDMI-A-1:1280x720@60D/' /boot/firmware/cmdline.txt
sudo reboot
```

Lessons

- Before flashing, check that the key shown in Imager ends the same as type C: \Users\Venkata\.ssh\id_ed25519.pub. Imager remembers old keys.
- Always back up .ssh before changes.
- With password login off, losing id_ed25519 means re-flashing or using a keyboard and screen.
- No passwords or key contents are stored in the repo.

P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

Summary

- Read Master Specification v1.0 and pulled out the P0 (Foundation) scope, done-criteria and phase gate.
- Wrote `docs/P0-readiness-checklist.md`: hardware, software with versions and links, Pi flashing steps, accounts.
- Installed and verified the developer tools on the Windows PC (all checksums matched the official ones).
- Flashed the Pi. The first attempts failed on SSH keys (including the private key being pasted into Imager); the key was replaced and SSH worked.
- Wrote the hub setup (`hub/scripts/setup-hub.sh` + `hub/system/*`) and ran it on the Pi at `192.168.1.241` (TELUS network). HTTPS worked by name and IP.
- Wrote `docs/P0-mini-spec.md` (P0 build order, decisions DL-21 to DL-25).
- Decisions from Sreya: RTC skipped for now; cloud backup tested later when client IDs exist; test phones are Android 9 and iPhone 13 Pro (iOS 26.6.1); update channel option A (source stays private, signed packages in a separate public repo).

1. Spec review

Source: `SDLC Docs Tools software installers and ReadMe/MASTER Specifications/Chedam - Master Specification v1.0.pdf` (41 pages).

The PDF text was extracted with `pdftotext -layout`. The release-plan and architecture pages are images, so they were rendered with PyMuPDF and read as pictures.

P0 requirements: FR-1.01 to 1.10, 1.13, 1.14, 1.15; FR-12.01 to 12.04, 12.08, 12.09.

Done when: 3 devices log in, the hub survives a power pull, an update installs and rolls back. The phase gate adds: memory under 150 MB (NFR-08), a load test (R6), an iOS/Android HTTPS camera test (R5), and a verified backup and restore.

2. Developer PC tools

Already present: Git 2.52.0, VS Code 1.135, Python 3.13, Windows OpenSSH.

Tool	Version	Verified against	Result
Node.js	24.21.0 (LTS Krypton)	nodejs.org SHASUMS256 + OpenJS signature	Match, installed
PocketBase (Windows)	0.40.4	PocketBase <code>checksums.txt</code>	Match
DuckDB CLI	1.5.6	GitHub release digest	Match
Raspberry Pi Imager	2.0.11.1	GitHub digest + Raspberry Pi Ltd signature	Match, installed
DB Browser for SQLite	3.13.1	winget manifest + SignPath signature	Match, installed
mkcert	1.4.4	Scoop manifest (file is unsigned)	Match
GitHub CLI	2.102.0	n/a	Installed, logged in as <code>repatikosamid-prog</code>

Commands used for verification (Git Bash):

```
sha256sum *.msi *.exe *.zip
curl -s https://nodejs.org/dist/v24.21.0/SHASUMS256.txt | grep x64.msi
```

```
powershell Get-AuthenticodeSignature <file>
```

Failed case: mkcert not recognized

- Symptom: `mkcert -install` → "The term 'mkcert' is not recognized".
- Cause: the file was still named `mkcert-v1.4.4-windows-amd64.exe`, and `C:\Tools` was not on PATH. PowerShell also never runs a program from the current folder without `.\`.
- Fix: copied `mkcert.exe`, `pocketbase.exe` and `duckdb.exe` to the top of `C:\Tools` and added `C:\Tools` to the user PATH. A new PowerShell window was needed for the change to show.

Then Sreya ran `mkcert -install` (local dev CA added to the Windows trust store) and `gh auth login` (fine-grained token).

The installers folder was added to `.gitignore`. PocketBase had created test data there.

3. Flashing the Pi (first time)

Imager settings: Pi Zero 2 W, Raspberry Pi OS Lite 64-bit, hostname `chedam`, user `chedam`, Wi-Fi 2.4 GHz, country CA, time zone America/Vancouver, SSH public-key only.

Failed case 1: hostname and username

- The hostname was typed as `Chedam` and the username as the GitHub name.
- Fix: lowercase `chedam` for both, so every script and guide matches.

Failed case 2: no SSH key existed

- SSH was set to password authentication, and the PC had no key yet.
- Fix: `ssh-keygen -t ed25519 -C "chedam-dev"`.

Failed case 3: Permission denied (publickey)

- Debug output showed the PC offering its key and the Pi refusing it for both usernames:

```
ssh -v -o BatchMode=yes chedam@chedam.local true
```

- Cause: the key written to the card was not the PC's key.
- The "host key differs" warning came from an old `192.168.1.241` entry in `known_hosts`. It was removed with `ssh-keygen -R`.

Failed case 4 (security): private key pasted into Imager

- Imager listed 7 "keys". They were the 7 lines of the **private** key (`BEGIN OPENSSH PRIVATE KEY`).
- Risk: the private key was copied onto the SD card and partly shown in a screenshot.
- Fix: removed all 7 entries, deleted the key pair, generated a new one, and pasted **only** `id_ed25519.pub` (one line, `ssh-ed25519 ... chedam-dev`). Re-flashing overwrote the card.

Failed case 5: `.local` name not resolving from Windows tools

- Git Bash ssh and Windows ssh both failed on `chedam.local`, while `Resolve-DnsName` worked.
- Fix: an SSH alias `chedam` pointing at the IP, in `~/.ssh/config`. The Pi first booted slowly (~5 min), so early checks also failed.

Login worked: `chedam@chedam`, Debian 13 trixie, aarch64, 415 MB RAM, 117 GB card.

Admin access

`sudo` asked for a password, so Sreya ran this once:

```
ssh -t chedam "echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmo
```

4. Hub setup (first time, Pi at 192.168.1.241)

OS update (ran in the background, ~15 min):

```
sudo apt-get update && sudo apt-get -y full-upgrade && sudo apt-get -y install chrony sqlite3 ufw
```

PocketBase 0.40.4 linux_arm64 was downloaded on the Pi and checked with `sha256sum -c` against `checksums.txt`.

Files written in the repo:

File	Purpose
hub/scripts/setup-hub.sh	Idempotent setup: packages, service user <code>chedam-hub</code> , <code>/opt/chedam</code> , PocketBase service, Caddy, watchdog, journald, Wi-Fi power save off, boot config, firewall
hub/system/chedam-hub.service	PocketBase on 127.0.0.1:8090, auto-restart, <code>GOMEMLIMIT=100MiB</code> , <code>MemoryMax=220M</code> , hardening
hub/system/Caddyfile	HTTPS for <code>chedam.local</code> and the hub IP, using a per-store CA; plain HTTP only for <code>/ca.crt</code>
hub/system/watchdog.conf	Hardware watchdog 14 s
hub/system/journald.conf	Logs capped at 64 MB (SD card wear)
hub/system/wifi-powersave-off.conf	Lower Wi-Fi latency
hub/system/index.html	Placeholder page "Chedam hub is running"

Run on the Pi:

```
scp -r scripts system chedam:hub/  
ssh chedam 'sudo bash ~/hub/scripts/setup-hub.sh'  
ssh chedam 'sudo systemctl reboot'
```

Failed case 6: GitHub repo creation

- `gh repo create ... --public` → "Resource not accessible by personal access token".
- Cause: the fine-grained token cannot create repositories. Sreya has to create the public repo on github.com.

Failed case 7: HTTPS tests from the PC returned nothing

- Windows `curl` (Schannel) → "the revocation status is unknown". A private CA has no revocation server.
- Fix for command-line tests: `--ssl-no-revoke`. Browsers are not affected.

Failed case 8: Caddy error at start

- Caddy tried to install its root certificate into the Pi's own trust store and failed ("failed to execute sudo").
- Fix: the `skip_install_trust` global option in the Caddyfile.

Results after reboot

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon active
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.1.241/api/health</code> → 200
Certificate	Valid against "Chedam Store CA - 2026 ECC Root" (valid to 2036); IP cert has IP SAN
Memory	PocketBase 27 MB + Caddy 41 MB; total RAM rose from 415 to 462 MB after headless config
Watchdog	14 s
Temperature	52.6 °C, no throttling

5. Decisions recorded (docs/P0-mini-spec.md)

ID	Decision
DL-21	HTTPS via Caddy with an internal CA created on each hub (one per store); mkcert is dev-PC only
DL-22	Source repo private; signed update packages in a separate public repo
DL-23	RTC deferred for dev; required for the pilot
DL-24	Headless boot config (later changed to a switch, see 2026-10-05)
DL-25	PocketBase extended with JavaScript hooks; no Go build

Findings:

- Android 9 cannot resolve `.local`, so the hub needs a fixed IP.
- Android 9's Chrome no longer gets updates.
- Caddy leaf certificates last 12 hours, so the clock must be correct (RTC for the pilot).